AI Exam Answer Grader

A Reusable NLP Asset for Rubric-Based Descriptive Answer Assessment

Weeks 1 & 2 Deliverable — Project Proposal, Baseline & Initial Evaluation | Target: 12–18 September 2026

1. Project Proposal (1-page overview)

1.1 Team

Roll No.	Name	Group
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1.2 Problem Statement

Faculty spend a disproportionate amount of time manually evaluating descriptive (subjective) examination answers against a marking rubric. This is slow, inconsistent across evaluators, and does not scale with class size. There is a need for an NLP-based component that can assist — not replace — faculty by comparing a student's descriptive answer against a reference answer and rubric, and recommending a score with supporting feedback.

1.3 Institutional Users

- Faculty / course instructors — primary users, for first-pass grading assistance and workload reduction.
- Teaching assistants — for consistent, rubric-aligned pre-grading of large answer scripts.
- Examination cell / academic administration — for score audit, moderation and consistency checks across evaluators.
- Other AI University components — e.g., Group 1's Handwritten Examination Digitizer, which supplies structured question-wise answer text as input to this component.

1.4 Input

- Question text
- Reference / model answer
- Marking rubric (key concepts, weightage per concept)
- Student answer (digital text; structured JSON from Group 1 when available)
- Maximum marks for the question

1.5 Output

A structured, machine-readable assessment object, for example:

```
{ "question": "Q4", "score": 7, "maximum_score": 10, "feedback": "The answer correctly explains X but omits Y.", "confidence": 0.87 }
```

1.6 NLP Task Formulation

The core task is formulated as semantic textual similarity / short-answer scoring: given a reference answer (and rubric concepts) and a candidate student answer, predict a numeric score correlated with human-

assigned marks, and surface the rubric concepts that are present, partially present, or missing. This can be framed as (a) a regression/similarity-scoring problem over embeddings, and (b) an LLM-based rubric-grounded evaluation problem — the two approaches this group is required to compare.

1.7 Proposed Approach

- Approach A (Baseline): TF-IDF and sentence-embedding cosine similarity between student answer and reference answer, mapped to a score via a calibrated similarity-to-marks function.
- Approach B (Improved): LLM-assisted, rubric-grounded evaluation — the model is given the question, rubric concepts and student answer, and asked to identify which rubric concepts are covered before producing a score and feedback, rather than a single "grade this" prompt.
- Both approaches are evaluated against the same human-labelled test set so their agreement with human graders can be directly compared.

1.8 Dataset (planned)

- Primary source: our own class's mid-term/assignment descriptive answers (anonymized), collected via Teams, together with reference answers and rubrics prepared with faculty.
- Supplementary: publicly available short-answer grading datasets (e.g., Mohler dataset / SemEval short-answer scoring sets) for additional training/validation signal.
- All student-identifying information will be anonymized before use; no data will be uploaded to public AI services without anonymization.

1.9 Expected Institutional Value

A working prototype that gives faculty a rubric-aligned first-pass score and explainable feedback per answer, reducing grading turnaround time while keeping a human in the loop for final decisions. Structured, confidence-scored output allows the component to plug directly into an examination workflow and to consume Group 1's digitized answer output.

2. Alignment with Group 1 (Handwritten Examination Digitizer)

Per the assignment's cross-group integration requirement, this project is designed to consume Group 1's output format directly:

Group 1 Output Field	Consumed As	Group 2 Field
student_id	Record identifier for the assessment	(pass-through, not scored)
question	Used to select the matching reference answer & rubric	question
answer_text	Treated as the student answer to be graded	student_answer
confidence (OCR)	Low-OCR-confidence answers flagged before grading rather than auto-scored	upstream_confidence

This means our API contract (Section 4) accepts Group 1's JSON schema as a valid input shape, with the reference answer, rubric and maximum_score supplied by the instructor-side configuration rather than by Group 1.

3. Working Baseline (Approach A) — Status

3.1 Method Implemented

- Text preprocessing: lowercasing, punctuation/stopword handling, lemmatization.

- Vectorization: TF-IDF over reference and student answers; cosine similarity computed between the two vectors.
- Score mapping: similarity score linearly scaled to the question's maximum marks, with a simple concept-coverage check (keyword overlap against rubric key-terms) as a secondary signal.
- Output wrapped into the target JSON schema shown in Section 1.6, including a heuristic confidence estimate derived from similarity margin.

3.2 Initial Evaluation

A small pilot set of answers (drawn from our own coursework, human-marked by two of the group members acting as reference graders) was used for an initial sanity check of the baseline pipeline before the full labelled test set is built in Week 5.

Metric	Initial Pilot Result
Mean Absolute Error (MAE) vs. human marks	2.628 (on Mohler pilot set, n = 2,273; full labelled test set in Week 5)
Agreement within ±1 mark	11.7%
Pearson correlation with human scores	0.464 ($p < 0.001$)

Key finding: the largest scoring errors occur on questions with very short reference answers (e.g. "Through iteration."), where a correct student paraphrase using different vocabulary produces zero TF-IDF token overlap and a predicted score of 0 despite a human score of 5. This is a known limitation of pure lexical-overlap methods on short-answer grading and directly motivates Approach B (LLM/rubric-grounded evaluation) in Week 3–4. Note: the pipeline is functional end-to-end (rubric + reference + student answer → structured score) on the pilot set; full quantitative results with a proper labelled test set are scheduled for Week 5 (Evaluation & Error Analysis) as per the project timeline. This week's deliverable establishes the baseline is working and produces schema-valid, scaled outputs, which is the required Week 1–2 milestone.

4. Reusable Interface (planned contract)

Endpoint: POST /evaluate-answer — accepts question, reference_answer, rubric, student_answer, maximum_marks (optionally the full Group 1 JSON object for student_answer) and returns the structured score/feedback/confidence object shown in Section 1.6. Full OpenAPI-style documentation will be produced in Week 6 (Asset Engineering) per the project schedule.

5. Responsible AI Notes (preliminary)

- Advisory only: the component recommends a score; final marks remain a faculty decision (no fully automated grading).
- Privacy: all answers used for development/testing are anonymized before processing; no raw student-identifying data is sent to any external LLM API.
- Bias & hallucination: LLM-based feedback will be constrained to rubric concepts actually present in the source texts, and cross-checked against the baseline's similarity score to flag large disagreements for human review.

6. Next Steps (Week 3)

- Build the full labelled test set with human-assigned marks across a larger answer sample.
- Implement Approach B (LLM/rubric-grounded evaluation) as Version 1 of the improved pipeline.
- Run both approaches on the same test set to begin the baseline-vs-improved comparison due in Week 4.